# TNY285 Flyback SMPS Design Session – Detailed Engineering Notes

## 1. Project Objective

Design a mains-isolated offline flyback SMPS using TNY285 with professional-grade PCB layout practices. Key goals included isolation safety, EMI control, manufacturability, and portfolio-level documentation.

## 2. Schematic Completion

The schematic was fully completed, including rectification, bulk storage, flyback switch, RCD/Z clamp, optocoupler feedback, Y-capacitor, auxiliary supply, and output filtering.

## 3. Stackup Selection

A 4-layer PCB stackup was selected to enable controlled return paths, plane isolation, EMI suppression, and robust creepage enforcement.

## 4. Net Class Definition

Three primary net classes were created: PRIMARY_HV, DRAIN_HOT, and SECONDARY_LV.

## 5. Clearance and Creepage Rules

Explicit clearance rules were defined between PRIMARY_HV and SECONDARY_LV nets with 10 mm spacing.

## 6. Polygon and Plane Rules

Polygon connect styles and plane clearance rules were defined per net class.

## 7. Component Placement Strategy

Critical HV loops were minimized; drain-hot copper was isolated; EMI parts preserved for future revs.

## 8. Isolation Geometry

A board cutout (isolation moat) was added beneath the optocoupler and Y-capacitor to increase creepage.

## 9. 3D Verification

Altium's 3D viewer was used to confirm physical spacing and creepage paths.

## 10. Routing Strategy

Routing deferred until rules locked; priority routing planned for HV and drain-hot loops.

## 11. End-of-Session Status

Schematic, BOM, placement, stackup, and rules finalized; routing scheduled next.

## 12. Engineering Significance

This session marked the transition from schematic to manufacturable PCB design.